

Assessment of Water Erosion Risks in the Oued Beht Watershed, Upstream of the Ouljet Essoltane Dam: A Combined Approach Using the Erosion Potential Model and PAP/CAR Standards

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Abstract

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Morocco's semi-arid environment, diverse geography, and extensive agricultural practices present serious water erosion problems for the Ouljet Essoltane watershed, a sub-basin of the Beht River. This study evaluates the qualitative and quantitative risks of water erosion using the Erosion Potential Model and the PAP/CAR standards. According to the PAP/CAR technique, high erosion classes make up 11% of the watershed, whereas low to very low erosion classes predominate, making up roughly 73%. The predominant erosion forms (sheet, rill, and gully erosion) collectively affect 76% of the area within the Ouljet Essoltane watershed. The EPM results indicate that medium to high erosion levels, primarily in the central and upper zones, covers 76% of the watershed. The annual soil loss in an average year was calculated as 1600 m³ /km²/year. This analysis underscores the urgent need for targeted soil conservation practices tailored to the specific conditions of each erosion class within the watershed, to ensure sustainable land systems in semi-arid areas and to reduce the risk of erosion in highly vulnerable regions.

Keywords: Water erosion; Soil protection; PAP/CAR model; EPM model; Ouljet Essoltane watershed; Morocco

1. Introduction

Water erosion constitutes one of the most common environmental challenges in semi-arid regions resulting in deposition in water bodies; a reduction in agricultural productivity and the soil quality (Brahim et al., 2020; Benzougagh et al., 2024). Morocco experiences an extended dry period characterized by irregular rainfall, and the Beht River basin, which includes the Ouljet Essoltane watershed, is severely affected by these climatic conditions (Moutaoikil et al., 2024; Lakhili et al., 2025). Ouljet Essoltane watershed, with a perimeter of approximately 414 km and an area of 4407 km² is one of the most fertile lands in terms of agriculture, and thus, the evaluation of soil erosion hazards of the area is essential for sustainable land management. Human, topographic and climatic factors all affect

soil erosion in semi-arid regions. The losses are exaggerated by heavy rainfall on steep slopes, especially after vegetation loss, which accelerates erosion due to unsustainable agricultural practices (Bammou et al., 2024; Moutaoikil et al., 2024). The relationship between these factors has led to significant land degradation in soils of the Ouljet Essoltane watershed compromising the quality of water and the productivity of crops. The objective of this research is to assess the risk of water erosion in the watershed of Ouljet Essoltane watershed by combining two complementary methods, and so, the Erosion Potential Model (EPM) for qualitative and quantitative analysis and the PAP/CAR approach qualitative analysis. EPM groups the soil loss according to topographical, climatic and land use variables, while PAP/CAR models, makes use of predictive and descriptive mechanisms to determine the category of erosion. By integrating a number of methods, this study provides a comprehensive overview of erosion dynamics and contributes to the development of targeted soil conservation strategies.

2. Materials and Methods

2.1. Study Area

The Ouljet Essoltane basin is part of the Beht River basin area in Morocco (southwest of the Sebou basin). The watershed experiences a Mediterranean climate with semi-arid characteristics, It spans an area of 4,407 km² with a perimeter of approximately 414 km (Fig.1) (Lakhili et al., 2018). The excess water sustains surface runoff and groundwater recharge, with an annual rainfall of around 515.76 mm and an estimated evapotranspiration of 337.9 mm. The topography ranges from gentle slopes in the lower reaches to steeper gradients in the upper and middle sections, influencing erosion patterns. Land use is predominantly agricultural, making the watershed more susceptible to erosion (El Aoufir et al., 2024).

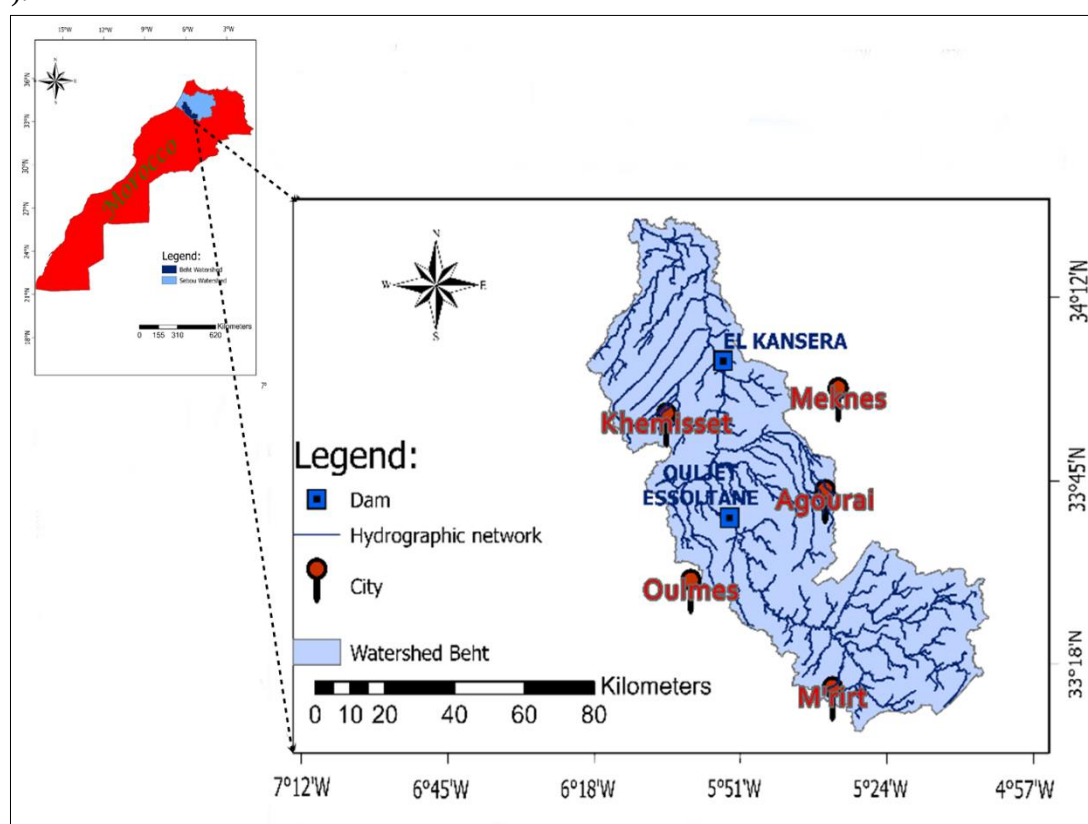


Fig. 1. Situation map of the study area showing the Oued Beht watershed and major hydrological features

2.2. Geological Context

Geologically, the Ouljet Essoltane basin is underlain by a complex stratigraphy that includes formations from the Paleozoic era to the Quaternary. The basin features sedimentary rocks, including shales, sandstones, and limestones, which have been shaped by tectonic activities over millions of years. These geological formations play a crucial role in determining soil types and their erodibility, as well as influencing the hydrological behavior of the area. The geological makeup also affects the stability of slopes within the basin, with certain formations being more prone to erosion than others. For instance, the presence of clay-rich soils, which are common in some areas, can lead to increased runoff and higher rates of soil erosion during heavy rainfall events.

2.3. Data Collection

Field surveys, satellite images (Landsat and Sentinel-2), and topographic maps (1:100,000) were used to collect information from November 2013 to June 2015. Topographic maps, including slope and hypsometric maps, were generated from DEMs at 30m spatial resolution. Cover and land use were recorded by remote sensing and validated by ground-truth observations. The temperature and rainfall records were obtained from El Kansera and Ouljet Soltane weather stations.

2.4. Methodologies

2.4.1. PAP/CAR guidelines

The PAP/CAR recommendations provide a qualitative framework for assessing erosion risk through both descriptive and predictive stages (Labaci et al., 2020; Laaouidi et al., 2025). Several tasks are employed during the prediction phase. Erodibility Map: This map is developed by combining categories of rock data with slopes (obtained from digital elevation models) to show soil sensitivity as a function of geological composition. Vegetation Protection Map: This map uses data on land use and land cover to assess the protective role of vegetation in preventing erosion. Combined Erodibility and Soil Protection Maps: This map categorizes low, medium and high-risk areas with soil erosion levels (Fig. 2). During the field campaign, field work and aerial image interpretation were performed which served to map different erosion types with sheets, waterways and valleys. The combination of the descriptive and predictive steps generates a full erosion map that shows the current situation and the erosion susceptibility (Armsworth et al., 2025).

2.4.2. Erosion potential model (EPM)

The Erosion Potential Method (EPM), developed by Gavrilovic, is used to assess soil erosion, (Eq.1). Taking into account landform, land use, precipitation and soil condition, this formula was developed to calculate the potential rate of soil erosion. These factors are integrated to estimate potential soil loss in a given area. This model may be used to better understand and manage the threats of soil erosion:

$$W = T \cdot H \cdot \pi \cdot Z^{1.5}, \quad (1)$$

Where: W: Annual soil loss (m³/km²/year), T: Temperature coefficient, based on average annual temperature, H: Precipitation coefficient, derived from annual rainfall, Z: Erosion coefficient that incorporates vegetation cover, slope, and soil erodibility.

In the watershed, the model was applied to estimate the soil loss rates and map the erosion coefficients using GIS (El Brahimi et al., 2024; Trendafilov et al., 2025). The parameters were validated by field measurements and integrated with land use maps, DEMs and climate data (Fig. 2).

2.5. Data Analysis

Among the geographic data processing methods conducted using GIS were vegetation cover mapping, slope analysis, and lithofacies classification (ArcGIS 10.8). R statistically analyzed correlations between erosion metrics. Both PAP/CAR and EPM results were calibrated and validated using field data.

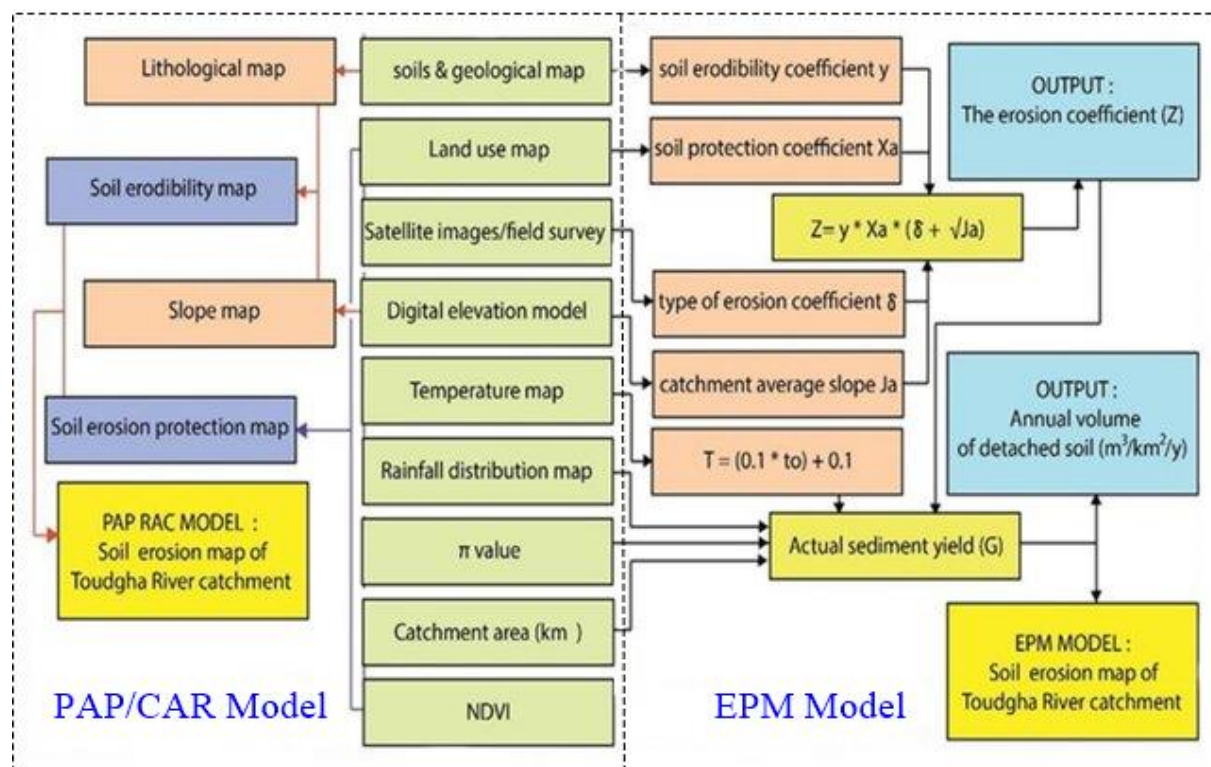


Fig. 2. Techniques used to run the PAP/RAC and EPM models on GIS software (Elbadaoui et al., 2023)

3. Results

3.1. Qualitative Erosion Assessment (PAP/CAR)

The PAP/CAR method classified 73% of the Ouljet Essoltane watershed as having low to very low erosion risk, primarily in the southwestern and downstream areas, which are characterized by gentle slopes and dense vegetation cover. High erosion risk areas, accounting for 11%, were concentrated in the central-western and downstream regions, considered by steeper slopes and sparse Vegetation. The most prevalent erosion forms were sheet, rill, and gully erosion, collectively occupying 76% of the watershed area. Stable land, with very low erosion risk, covered 24% of the watershed (Table 1). The consolidated erosion map (Fig. 3) highlighted high-risk zones associated with rill and gully erosion, particularly in areas with areas of intensive agricultural activity and exposed soils.

Table 1. Superficies of Erosion Classes in the Ouljet Essoltane Watershed (PAP/CAR Method)

Erosion Class	Superficie (%)	Area (km ²)
Very Low	40	1762.8
Low	33	1454.3
Medium	16	705.1
High	11	484.8
Stable Land	24	1057.68

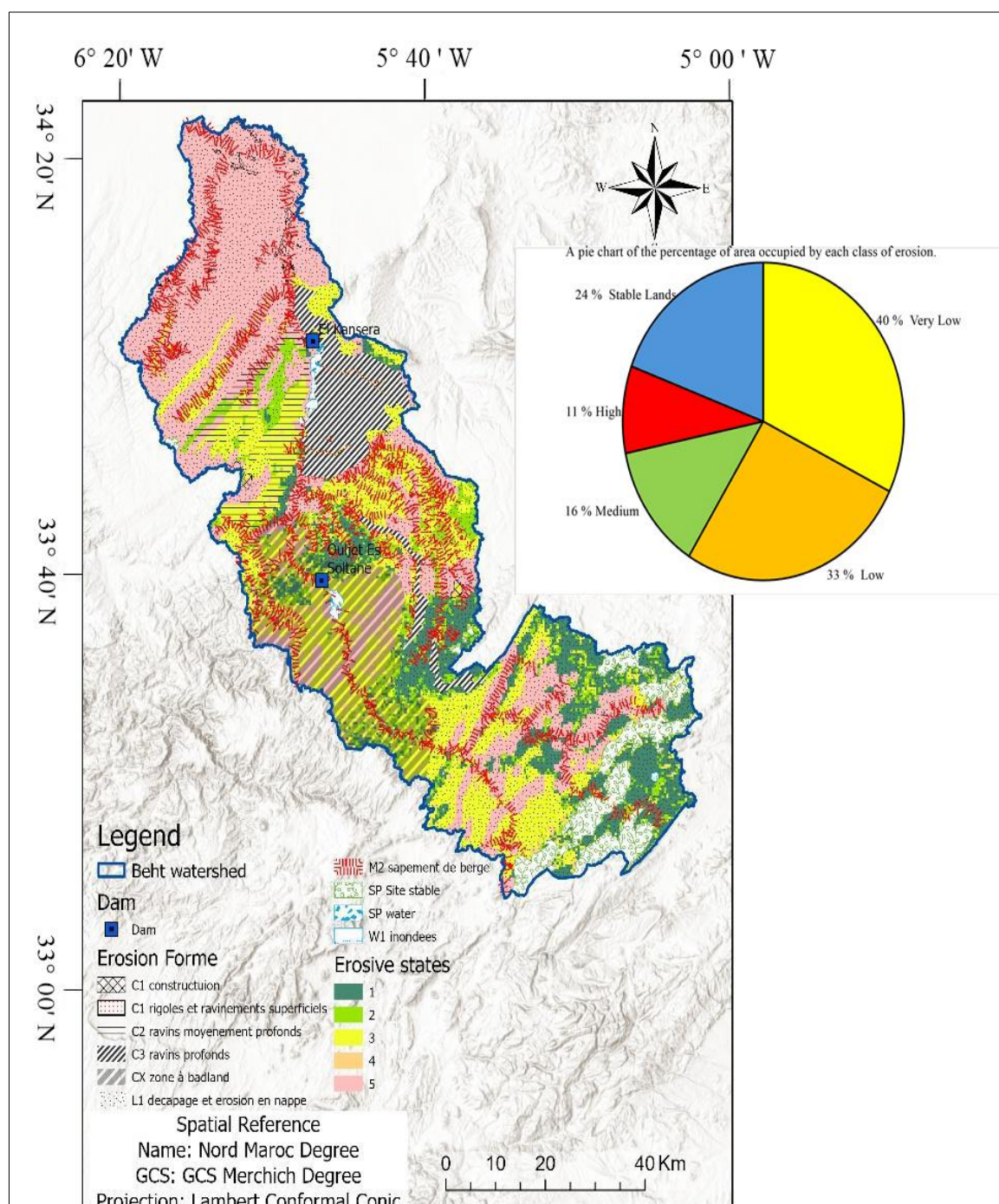


Fig. 3. Consolidated PAP/CAR Erosion Map of the Ouljet Essoltane Watershed

3.2. Quantitative Erosion Assessment (EPM)

The Erosion Potential Model (EPM) has provided valuable insights into the soil erosion dynamics within the Oued Beht watershed, revealing an average annual soil loss of approximately 1600 m³/km²/year. The spatial distribution of erosion risk across the watershed is illustrated in Fig. 4, which categorizes the erosion into three distinct classes based on annual soil loss: low, medium, and high erosion zones. Each classification reflects varying levels of soil degradation, significantly influencing land management and conservation efforts in the region.

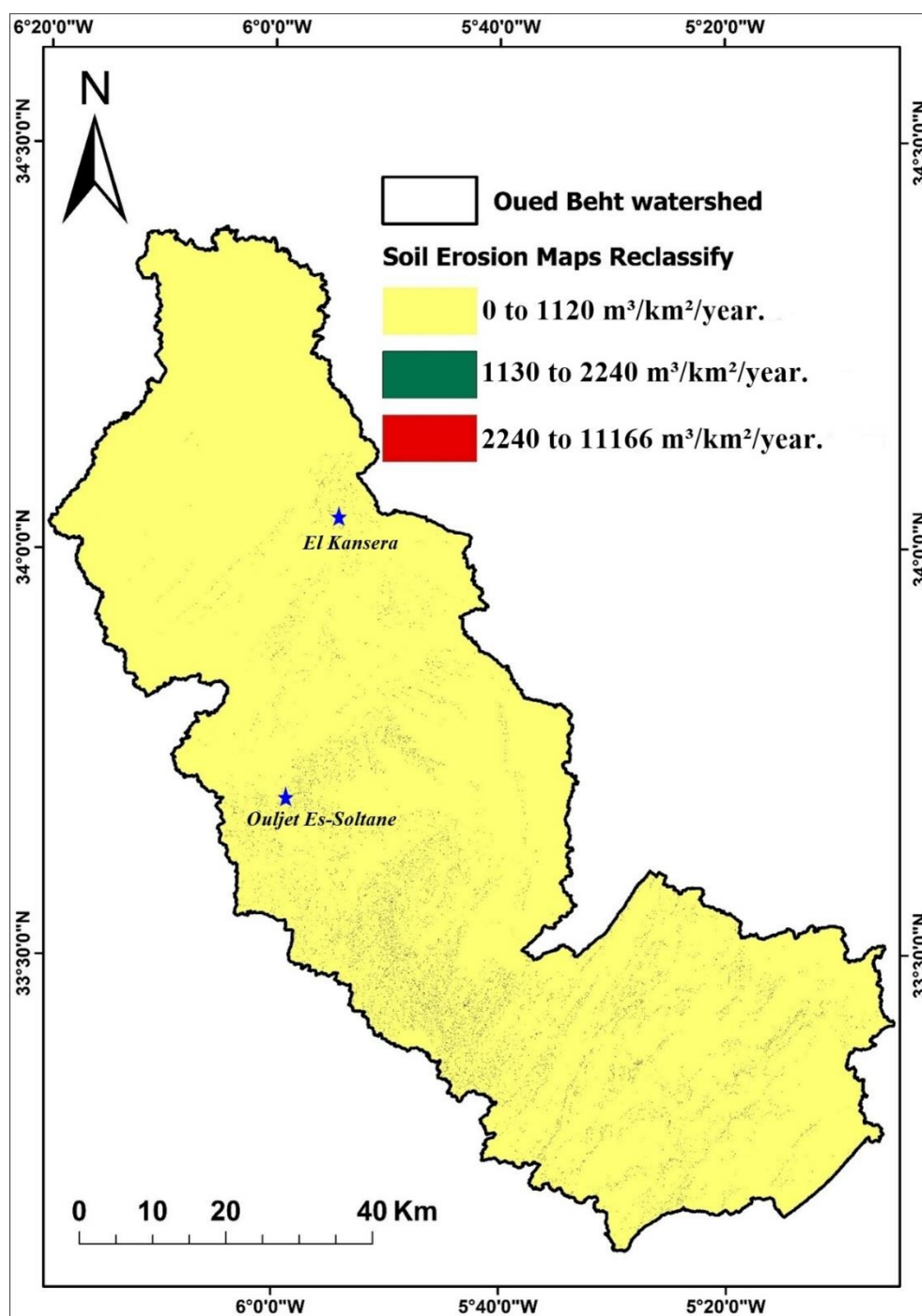


Fig. 4. EPM Soil Loss Map of the Ouljet Essoltane Watershed

The low erosion class, which encompasses soil loss from 0 to 1120 m³/km²/year and is depicted in yellow on the map, represents areas that are generally well-protected by vegetation and exhibit minimal soil disturbance. These regions are vital for maintaining soil health and biodiversity, serving as critical buffers against erosion. They are likely to support sustainable agricultural practices, making their preservation essential. This class dominates the watershed, covering an estimated 70–75% of the total area and is primarily found in the northern and central sections.

In contrast, the medium erosion class, characterized by soil loss ranging from 1130 to 2240 m³/km²/year and marked in green, indicates moderate erosion levels. This category includes areas that may experience land-use pressures, such as agricultural activities that disturb the soil. While these regions are not critically degraded, they require careful management to prevent escalation into higher

erosion classes. Implementing practices such as crop rotation and cover cropping could enhance soil stability and mitigate further erosion.

The high erosion class, highlighted in red, identifies the most vulnerable areas within the watershed, with soil loss ranging from 2240 to 11166 m³/km²/year. These zones are experiencing significant soil loss, primarily due to anthropogenic activities like deforestation, overgrazing, and intensive agriculture. Immediate intervention is crucial in these regions to mitigate soil degradation. Strategies may include reforestation, the construction of erosion control structures, and the adoption of improved land-use practices to restore soil health.

The observed spatial pattern underscores the critical roles of topography and land cover in controlling erosion intensity within the Oued Beht watershed. The combination of steep slopes and inadequate vegetation in certain areas leads to heightened erosion risks, while more stable regions benefit from gentle topography and robust plant cover. Effective land management strategies should focus on preserving and enhancing vegetation in vulnerable areas to mitigate soil degradation and maintain watershed health.

In conclusion, the analysis of the soil erosion map for the Oued Beht watershed underscores the need for targeted management interventions tailored to the specific conditions of each erosion class. By addressing the causes of erosion in medium and high-risk areas while preserving low erosion zones, sustainable land management practices can be developed to protect soil resources and enhance agricultural productivity. The comparison of results between PAP/CAR and EPM demonstrated a strong correlation, affirming the EPM's reliability in pinpointing high-risk zones (Table 2).

Table 2. EPM Soil Loss Estimates in the Ouljet Essoltane Watershed

Erosion Category	Soil Loss (m ³ /km ² /year)	Superficie (%)
Very Low	0.12–500	24
Medium	500–5000	60
High	5000–20400	16

4. Discussion

The results obtained from this qualitative erosion study using the PAP/CAR method are consistent with other research on erosion conducted in the Middle Atlas region, particularly within the Beht watershed (El Gaatib et al., 2015; Mimich et al., 2018; Ait Yacine et al., 2019; Lakhili et al., 2025). Studies ranging from the Tabular Middle Atlas (Ousmana et al., 2017) to the Allal El Fassi dam watershed in the eastern Middle Atlas (Maktite and Faleh, 2017), as well as those in the central Middle Atlas (Limame et al., 2024), highlight the effectiveness of this approach for estimating water erosion. These studies consistently reveal the predominance of sheet erosion, followed by rill and gully erosion, landslide erosion, and, finally, badlands erosion.

The dominance of low to very low erosion classes (73%) in the Ouljet Essoltane watershed reflects the stabilizing effect of vegetation cover and gentle slopes in the southwestern and downstream areas. However, the 11% of high erosion risk areas, characterized by rill and gully erosion, indicate significant soil degradation in the middle and downstream regions. As already found in other semi-arid catchments, these processes become more pronounced in portions of greater slope, sparser vegetation as well as areas with intensive agricultural practices (Corrao et al., 2017; Dofee et al., 2024; Yono et al., 2025). With a mean soil loss of 1600 m³/km²/year, the EPM mean results provide a comprehensive understanding of erosion. These results indicate that the watershed experiences moderate to high levels of erosion. 76% of the central and upper sites exhibited moderate to high erosion rates, consistent with the PAP/CAR results, highlighting the influence of topography and land use on erosion dynamics. Agriculture,

including plowed fields and over-grazed states, facilitates soil erosion, as they expose the land to rainfall runoff and reduce vegetation cover (Bilotta et al., 2007; Musa et al., 2024; Jothimani et al., 2025).

The efficacy of combining qualitative and quantitative techniques is evidenced by the agreement between the PAP/CAR and EPM results (Oudjane et al., 2021; Ed-Daoudy et al., 2024; Baiddah et al., 2025). The PAP/CAR approach effectively identifies erosion hotspots and spatial erosion trends, while, EPM approach ensures reliable detection of erosion and facilitates efficient allocation of conservation resources (Elbadaoui et al., 2023; Limame et al., 2024). Using GIS and field data parameters in both methods can improve their accuracy and allow for a more comprehensive erosion assessment (Benzougagh et al., 2022; Youssef et al., 2023). Focused soil conservation measures are indispensable in intermediate and downstream reaches, where high-risk erosion zones have already been detected. Measures that are Measures likely to control soil erosion such as contour plowing, terracing or reforestation, can stabilize the soil and reduce runoff in hilly regions. Promoting sustainable farming practices such as planting cover crops and using reduced tillage can enhance soil coverage and reduce the risk of erosion. Such approaches would also complement global efforts to combat land degradation in the semi-arid regions (Abdullahi et al., 2023; Yan et al., 2024; Chakraborty et al., 2025).

There are however limitations of the PAP/CAR and EPM techniques although they provide meaningful insights. These include the need for higher-resolution digital elevation models to accurately capture fine-scale erosion patterns (Yang et al., 2024; Colacicco et al., 2025) and their reliance on historical analysis (Lamane et al., 2022; Ed-Daoudy et al., 2024). For future work, researchers need to focus on monitoring the changes in erosion with time and how effectively conservation is succeeding. Climate change predictions, especially related to alterations in rainfall intensity, can be considered to enhance estimations of erosion risk.

The study found that the Erosion Potential Model (EPM) estimated an average soil loss of approximately 1600 m³/km²/year for the Ouljet Essoltane watershed. In contrast, the RUSLE model produced an average soil loss of 22.5 t/ha/year. This comparison highlights a notable difference, as the EPM results suggest a lower soil loss compared to the RUSLE estimates when converted to a similar scale. Specifically, 22.5 t/ha/year corresponds to approximately 2250 m³/km²/year, indicating that the EPM's estimates are somewhat lower than those from RUSLE.

The differences in results may arise from the methodologies employed by each model. The EPM relies on factors such as topography, land use, and historical erosion data to estimate potential erosion, focusing mainly on physical landscape characteristics. In contrast, RUSLE incorporates quantitative parameters such as rainfall erosivity, soil erodibility, slope length, and land cover management, which may lead to variations in the predicted soil loss. This analysis underscores the importance of using multiple models to assess soil erosion, as each provides unique insights into the erosion dynamics of the watershed. It also highlights the need for further validation and calibration of these models to enhance their predictive accuracy in the study area.

Based on the results of this study, as well as the findings of previous studies conducted in the watershed, it is clear that practical erosion control measures are needed to mitigate runoff and soil loss in vulnerable areas. To address these issues, we propose the following operational measures: (i) Establishing vegetative buffer strips along the banks of watercourses can help reduce runoff speed and promote sediment deposition. Native vegetation can stabilize soil and enhance biodiversity. (ii) Implementing contour farming and terracing in agricultural areas can significantly reduce slope length and consequently lower erosion rates. These practices help retain soil moisture and improve fertility. (iii) Installing check dams or silt traps in gullied areas can capture sediment before it enters the reservoir. These structures can slow water flow and promote sediment settling. (iv) Planting trees and restoring natural vegetation in critical areas can enhance soil stability and reduce erosion. Forest cover also plays a vital role in maintaining the hydrological balance of the watershed. (v) Encouraging farmers to adopt soil conservation practices, such as reduced tillage and cover cropping, can help maintain soil structure

and reduce susceptibility to erosion. (vi) Raising awareness among local communities about the impacts of erosion and the importance of conservation practices can foster community involvement in protecting the watershed.

By implementing these measures, we can significantly enhance the resilience of the Ouljet Essoltane dam against siltation, thereby extending its operational lifespan. These operational strategies are crucial not only for mitigating the immediate impacts of soil erosion but also for ensuring the long-term sustainability of the dam's functionality.

5. Conclusions

The comprehensive analysis of water erosion risk in the Ouljet Essoltane watershed provides significant new insights into the process of soil degradation within the Moroccan Beht River basin. In addition to estimating erosion risks, we have located critical vulnerable areas for soil loss through the EPM and in accordance with the technical guidelines of PAP/CAR. Results indicate that although 73% of the watershed has low to very low erosion risk, the remaining 11% classified as high-risk zones requires urgent intervention due to its crucial role in supporting agricultural productivity and maintaining water quality.

The relatively high average annual erosion rate of about 1,600 m³/km²/year, particularly in the central and upper sub-basins, underscores the necessity of targeted soil management strategies. Because of the complexity of the erosion processes in semi-arid regions, affected by geological and climatic settings and anthropogenic factors, integrated management approaches are required. Long-term monitoring and higher-resolution datasets are needed in future studies to better understand erosion dynamics and to assess the effectiveness of existing conservation measures. Climate change is also influencing erosion patterns, and its effects should be integrated into adaptive and sustainable land management strategies.

This research contributes to the existing literature on soil erosion in semi-arid environments and provides valuable references for decision makers and land managers aiming to reduce erosion vulnerability in at-risk watersheds. The implementation of the recommended soil conservation measures is essential to promote sustainable agricultural practices and enhance the resilience of the Ouljet Essoltane watershed against water erosion.

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